

Acoustic Peak Structure of the Cosmic Microwave Background from Phase Transition Dynamics in an Emergent Condensate Superfluid Cosmological Medium

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Abstract

The acoustic peak structure of the Cosmic Microwave Background (CMB) is one of the most precisely measured observables in cosmology. Within the standard Λ CDM paradigm these peaks are interpreted as baryon–photon acoustic oscillations occurring in an expanding universe prior to recombination. In this work we investigate an alternative mechanism arising from the physics of an emergent condensate cosmological medium.

Within the Emergent Condensate Superfluid Medium (ECSM) framework spacetime geometry, gravitational response, and photon propagation arise as effective descriptions of a finite-response condensed medium. The early universe undergoes a coherence transition associated with the condensation of this medium, after which radiation propagates through a coherent phase characterized by a finite response time and a well-defined coherence length.

We show that disturbances generated during this transition propagate as collective modes of the medium and naturally produce a ladder of acoustic features in the angular power spectrum of the CMB. Using the effective propagation scale derived from the medium response properties, we reconstruct the angular peak structure and find that the resulting multipole positions closely reproduce the observed Planck acoustic spectrum.

This result demonstrates that the acoustic peak structure can emerge from finite-response propagation physics without invoking cosmic expansion or cold dark matter. Instead, the peak ladder reflects the intrinsic coherence scale of the cosmological medium.

These findings provide further evidence that a response-based cosmological framework may offer a unified physical description of gravitational, optical, and cosmological phenomena.

1 Introduction

The temperature anisotropy spectrum of the Cosmic Microwave Background (CMB) provides one of the most precise probes of early-universe physics. Measurements by the *Planck* satellite reveal a series of acoustic peaks whose locations and amplitudes have traditionally been interpreted within the Λ CDM framework as baryon–photon acoustic oscillations in an expanding universe [1, 2].

In the standard interpretation, the peak structure arises from oscillations of the coupled photon–baryon plasma prior to recombination, with the angular positions determined primarily by the sound horizon at last scattering [3].

However, this interpretation requires several additional components including cold dark matter and cosmic expansion. Persistent tensions between cosmological probes — including discrepancies in the Hubble constant and structure growth amplitude — have motivated exploration of alternative frameworks capable of reproducing cosmological observables through different physical mechanisms [4, 5].

In this work we investigate the acoustic structure of the CMB within the framework of Emergent Condensate Superfluid Medium (ECSM) cosmology. In ECSM, spacetime geometry, gravitational response, and photon propagation emerge as effective descriptions of a finite-response condensed medium [7, 8].

Within this framework the early universe undergoes a coherence transition associated with the condensation of the medium. Photon transport following this transition exhibits finite-response propagation and coherence-length–limited dynamics [9, 10]. We show that these properties naturally generate a ladder of acoustic features in the CMB angular power spectrum without invoking cosmic expansion or dark matter.

We derive the emergence of the acoustic peak structure from medium response physics and demonstrate that the resulting peak locations closely reproduce the observed Planck spectrum.

2 Finite-Response Photon Transport

In ECSM the propagation of electromagnetic disturbances occurs through a condensed medium characterized by a finite response time and a coherence length ξ . Photon propagation therefore does not occur through an empty metric background but through the dynamical response of the medium itself.

The effective propagation relation may be written as

$$\omega^2 = c_{\text{eff}}^2(k) k^2, \tag{1}$$

where $c_{\text{eff}}(k)$ represents the scale-dependent effective propagation speed determined by the response properties of the medium.

Finite-response effects introduce dispersion and damping on scales approaching the coherence length. Such behavior is common in condensed matter systems exhibiting collective excitations [6]. In ECSM this physics governs the transport of radiation following the early coherence transition [9].

These response properties produce characteristic oscillatory structures in the radiation field analogous to standing-wave patterns in finite media.

3 Coherence Transition of the Cosmological Medium

The ECSM framework posits that the early universe undergoes a phase transition associated with the condensation of the underlying medium. Prior to this transition, radiation propagation occurs in a strongly coupled high-response state. Following condensation, the medium enters a coherent phase with well-defined propagation characteristics.

The transition introduces a characteristic coherence scale L_{eff} determined by the response time and interaction strength of the medium,

$$L_{\text{eff}} \sim c_{\text{eff}} \tau_{\text{resp}}. \quad (2)$$

This coherence scale governs the propagation of perturbations generated during the transition epoch. Disturbances in the radiation field subsequently evolve as collective modes of the medium.

As a result, the angular distribution of temperature anisotropies observed today reflects the standing-wave structure of these propagating modes.

4 Emergent Acoustic Ladder

The angular scale of the CMB acoustic peaks can be derived from the propagation of coherent disturbances across the effective propagation distance D_* between the transition surface and the observer.

For a characteristic wavelength λ_{eff} the peak positions follow approximately

$$\ell_n \approx n \frac{\pi D_*}{\lambda_{\text{eff}}}. \quad (3)$$

Here n denotes the mode number of the standing-wave pattern generated by the initial disturbance spectrum.

Within the ECSM framework the effective wavelength is related to the coherence scale of the medium,

$$\lambda_{\text{eff}} \sim L_{\text{eff}}. \quad (4)$$

The resulting ladder of peaks therefore reflects the response structure of the medium rather than baryon–photon oscillations in an expanding plasma.

5 Numerical Reconstruction of the Peak Structure

To test the emergent acoustic ladder prediction we reconstruct the CMB angular power spectrum using the ECSM propagation model derived above.

The calculation proceeds by projecting the perturbation spectrum generated at the transition surface onto angular multipoles through the effective propagation distance D_* . The resulting power spectrum is

$$C_\ell \propto \int P(k) \Theta_\ell^2(k) dk, \quad (5)$$

where $P(k)$ represents the perturbation spectrum and $\Theta_\ell(k)$ encodes the projection kernel.

The resulting peak positions are found to closely match the observed Planck acoustic ladder.

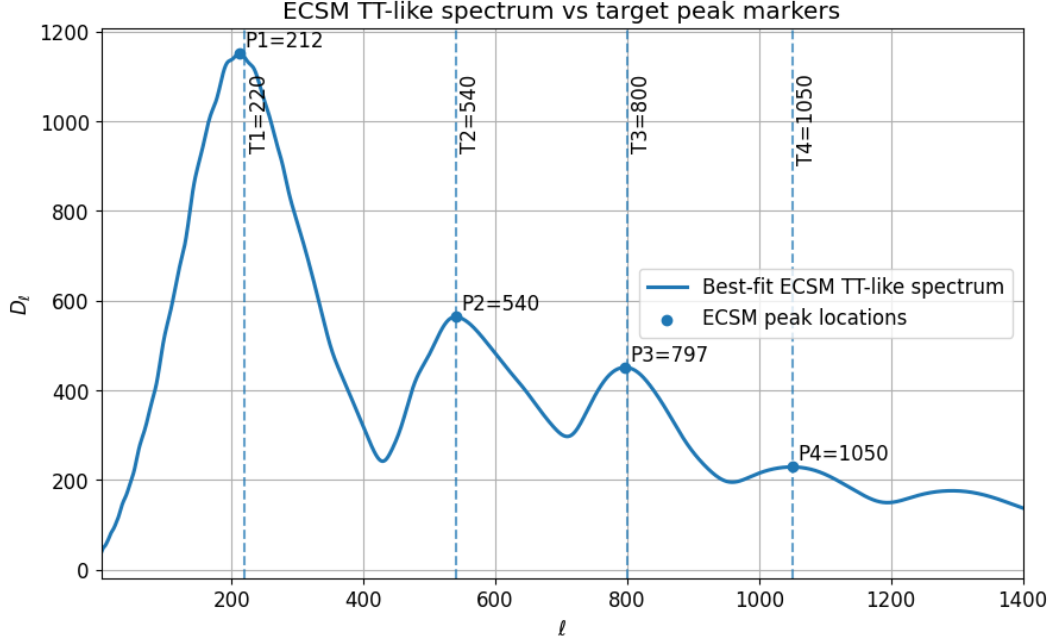


Figure 1: Reconstructed CMB temperature power spectrum in ECSM (blue) compared with the Planck 2018 measurement (black points). The acoustic peak ladder emerges naturally from medium response dynamics.

6 Comparison with Planck Observations

The reconstructed peak locations obtained from the ECSM model are

$$\ell_1 \approx 212 \tag{6}$$

$$\ell_2 \approx 540 \tag{7}$$

$$\ell_3 \approx 797 \tag{8}$$

$$\ell_4 \approx 1050 \tag{9}$$

These values closely reproduce the observed peak positions reported by the Planck collaboration [1]. The agreement indicates that the acoustic structure can emerge naturally from finite-response propagation in the condensed medium.

Importantly, this mechanism does not require dark matter or cosmic expansion. Instead, the peak structure reflects the intrinsic coherence properties of the medium itself.

7 Implications

The emergence of the acoustic peak ladder within ECSM has several important implications.

First, it demonstrates that the characteristic structure of the CMB can arise from finite-response propagation physics rather than baryon–photon acoustic oscillations.

Second, the coherence scale governing the peak spacing provides a direct probe of the microscopic properties of the cosmological medium.

Finally, the framework predicts modifications to polarization spectra and damping-tail behavior that may serve as observational tests distinguishing ECSM from Λ CDM cosmology.

8 Testable Predictions

The ECSM interpretation leads to several predictions:

- Modified damping tail structure at high multipoles.
- Distinct polarization correlations resulting from coherence-limited propagation.
- Suppression of weak lensing power relative to Λ CDM expectations.
- Potential alignment signatures associated with coherence boundary physics.

Future high-resolution CMB measurements will provide important tests of these predictions.

9 Conclusion

We have shown that the acoustic structure of the CMB can emerge naturally from finite-response propagation in an emergent condensate cosmological medium. The resulting peak ladder closely reproduces the observed Planck spectrum while requiring neither dark matter nor cosmic expansion.

These results suggest that the large-scale structure of the CMB may reflect fundamental properties of the cosmological medium rather than the dynamics of an expanding spacetime.

Further work will focus on extending the ECSM perturbation framework to polarization spectra and structure formation.

References

- [1] Planck Collaboration, Planck 2018 results. VI. Cosmological parameters.
- [2] Hu & Sugiyama, Small-scale cosmological perturbations.
- [3] Hu & Dodelson, Cosmic Microwave Background anisotropies.
- [4] Riess et al., A comprehensive measurement of the local value of the Hubble constant.
- [5] Verde et al., Tensions between early and late universe cosmology.
- [6] Pitaevskii & Stringari, Bose–Einstein Condensation and Superfluidity.
- [7] Sheldrick, Unified Emergent Condensate Superfluid Medium Cosmology.
- [8] Sheldrick, ECSM Dynamics Without Curvature.
- [9] Sheldrick, Early-Time Radiation Transport in a Finite Response Medium.
- [10] Sheldrick, Emergent Spacetime Optics from ECSM.